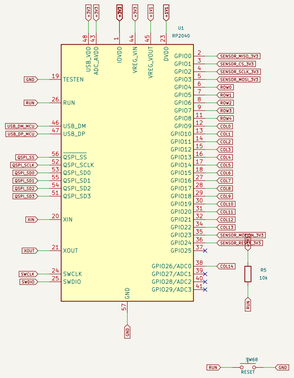
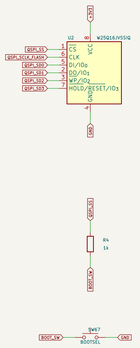
## 1 / RP2040 + FLASH + BOOT + RESET




SW60
RESET

SW67
BOOTSEL

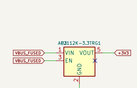
DECOUPLING: individual IOVDD / USB / ADC / DVDD pins; VREG 1uF input + output

C3 100nF
C4 100nF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C9 100nF
C10 100nF
C11 100nF
C12 100nF
C13 1uF
C14 1uF
C15 100nF

TP5 SWCLK / SWD
TP6 SWDIO / SWD
TP7 RUN / SWD
TP8 GND / SWD
TP9 +3V3 / SWD

## 2 / USB LOGICAL INPUT + 3V3 POWER

DRAFT: USB pads are individual logical nets, not JST pin order. CC/ESD/shield on existing daughterboard unverified.




R1 27R
R2 27R
R6 1k

Y1 ABM8-272-T3 12MHz CL10pF
C16 15pF
C17 15pF

TP1 VBUS_USB / USB LOGICALGND / USB LOGICAL
TP2 USB LOGICAL
TP3 USB LOGICAUSB_DP / USB LOGICAL
TP4 USB LOGICAL

F1 500mA hold / provisional
C1 1uF
C2 1uF

PWR_FLAG #FLG01
PWR_FLAG #FLG02

R3 10k
R4 1k
R7 27R
R5 10k
R8 10k

TP10 SENSOR_MISO_3V3
TP11 SENSOR_CS_3V3
TP12 SENSOR_SCLK_3V3
TP13 SENSOR_MOSI_3V3
TP14 SENSOR_MOTION_3V3
TP15 SENSOR_RESET_3V3
TP16 GND
TP17 +3V3

No module connector or 5V sensor connection. Verify revision and level translation before connection.

## 3 / CLOCK + SENSOR MCU-SIDE ONLY

## 4 / 5 ROW x 15 COLUMN MATRIX: COL -> switch -> diode ANODE -> CATHODE -> ROW (COL2ROW)

66 keys; IDs from preferred.keys; missing r1..3 c6..8 are intentional. Central 50x50mm is RESERVED, NOT A HOLE.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SW1 r0c0 Esc | SW2 r0c1 1 | SW3 r0c2 2 | SW4 r0c3 3 | SW5 r0c4 4 | SW6 r0c5 5 | SW7 r0c6 Vol- | SW8 r0c7 Mute | SW9 r0c8 Vol+ | SW10 r0c9 8 | SW11 r0c10 7 | SW12 r0c11 6 | SW13 r0c12 9 | SW14 r0c13 0 | SW15 r0c14 Bksp |
| D1 1N4148W | D2 1N4148W | D3 1N4148W | D4 1N4148W | D5 1N4148W | D6 1N4148W | D7 1N4148W | D8 1N4148W | D9 1N4148W | D10 1N4148W | D11 1N4148W | D12 1N4148W | D13 1N4148W | D14 1N4148W | D15 1N4148W |
| SW16 r1c0 Tab | SW17 r1c1 Q | SW18 r1c2 W | SW19 r1c3 E | SW20 r1c4 R | SW21 r1c5 T | | | | SW22 r1c9 Y | SW23 r1c10 U | SW24 r1c11 I | SW25 r1c12 O | SW26 r1c13 P | SW27 r1c14 \ |
| D16 1N4148W | D17 1N4148W | D18 1N4148W | D19 1N4148W | D20 1N4148W | D21 1N4148W | | | | D22 1N4148W | D23 1N4148W | D24 1N4148W | D25 1N4148W | D26 1N4148W | D27 1N4148W |
| SW28 r2c0 Caps | SW29 r2c1 A | SW30 r2c2 S | SW31 r2c3 D | SW32 r2c4 F | SW33 r2c5 G | | | | SW34 r2c9 H | SW35 r2c10 J | SW36 r2c11 K | SW37 r2c12 L | SW38 r2c13 ; | SW39 r2c14 Enter |
| D28 1N4148W | D29 1N4148W | D30 1N4148W | D31 1N4148W | D32 1N4148W | D33 1N4148W | | | | D34 1N4148W | D35 1N4148W | D36 1N4148W | D37 1N4148W | D38 1N4148W | D39 1N4148W |
| SW40 r3c0 Shift | SW41 r3c1 Z | SW42 r3c2 X | SW43 r3c3 C | SW44 r3c4 V | SW45 r3c5 B | | | | SW46 r3c9 N | SW47 r3c10 M | SW48 r3c11 , | SW49 r3c12 . | SW50 r3c13 / | SW51 r3c14 Shift |
| D40 1N4148W | D41 1N4148W | D42 1N4148W | D43 1N4148W | D44 1N4148W | D45 1N4148W | | | | D46 1N4148W | D47 1N4148W | D48 1N4148W | D49 1N4148W | D50 1N4148W | D51 1N4148W |
| SW52 r4c0 Ctrl | SW53 r4c1 GUI | SW54 r4c2 Alt | SW55 r4c3 Fn | SW56 r4c4 Space | SW57 r4c5 Del | SW58 r4c6 M1 | SW59 r4c7 M3 | SW60 r4c8 M2 | SW61 r4c9 Bksp | SW62 r4c10 Space | SW63 r4c11 Fn | SW64 r4c12 Left | SW65 r4c13 Down | SW66 r4c14 Right |
| D52 1N4148W | D53 1N4148W | D54 1N4148W | D55 1N4148W | D56 1N4148W | D57 1N4148W | D58 1N4148W | D59 1N4148W | D60 1N4148W | D61 1N4148W | D62 1N4148W | D63 1N4148W | D64 1N4148W | D65 1N4148W | D66 1N4148W |

NOT FOR FABRICATION — sensor and daughterboard interfaces provisional

Sheets: /
File: keycult66.kicad_sch

**Title: Keycult66 / electrical review draft**

| Size: A1 | Date: 2026-10-29 | Rev: DRAFT-A |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |